

```
!pip install chromadb
!pip install sentence-transformers
!pip install langchain
```

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```

```
import chromadb
from sentence_transformers import SentenceTransformer
```

```
client = chromadb.Client()

collection = client.create_collection(name="rag_collection")
print("Collection created successfully!")
```

Collection created successfully!

```
model = SentenceTransformer('all-MiniLM-L6-v2')
print("Model loaded!")
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
modules.json: 100% 349/349 [00:00<00:00, 27.5kB/s]

config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 8.28kB/s]

README.md: 10.5k/? [00:00<00:00, 596kB/s]

sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 4.77kB/s]

config.json: 100% 612/612 [00:00<00:00, 53.6kB/s]

model.safetensors: 100% 90.9M/90.9M [00:01<00:00, 453MB/s]

Loading weights: 100% 103/103 [00:00<00:00, 405.25it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key | Status | |
-----+-----+--
embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100% 350/350 [00:00<00:00, 11.6kB/s]

vocab.txt: 232k/? [00:00<00:00, 7.31MB/s]

tokenizer.json: 466k/? [00:00<00:00, 19.9MB/s]

special_tokens_map.json: 100% 112/112 [00:00<00:00, 9.97kB/s]

config.json: 100% 190/190 [00:00<00:00, 15.0kB/s]
Model loaded!

```

```

documents = [
    "Artificial Intelligence is the simulation of human intelligence.",
    "Machine Learning is a subset of AI.",
    "Deep Learning uses neural networks.",
    "Natural Language Processing deals with text data.",
    "RAG combines retrieval and generation.",
    "Neural networks are inspired by the human brain.",
    "Supervised learning uses labeled data.",
    "Unsupervised learning finds patterns in data.",
    "Reinforcement learning uses rewards and punishments.",
    "AI is widely used in healthcare and finance."
]

ids = [f"doc{i+1}" for i in range(len(documents))]

print("Dataset prepared!")

```

Dataset prepared!

```

embeddings = model.encode(documents)
print("Embeddings generated!")

```

Embeddings generated!

```

collection.add(
    documents=documents,
    embeddings=embeddings,
    ids=ids
)

print("Data stored in ChromaDB!")

```

Data stored in ChromaDB!

```

query = "What is Artificial Intelligence?"

query_embedding = model.encode([query])

results = collection.query(
    query_embeddings=query_embedding,
    n_results=3
)

```

Raw Results:

```
{'ids': [['doc1', 'doc2', 'doc10']], 'embeddings': None, 'documents': [['Artificial Intelligence is the simulation of human intelligence.'], ['Machine Learning is a subset of AI.'], ['AI is widely used in healthcare and finance.']]}
```

```
print("\nTop Retrieved Documents:\n")
```

```
for doc in results['documents'][0]:  
    print("-", doc)
```

Top Retrieved Documents:

- Artificial Intelligence is the simulation of human intelligence.
- Machine Learning is a subset of AI.
- AI is widely used in healthcare and finance.

```
def retrieve(query):  
    query_embedding = model.encode([query])  
    results = collection.query(  
        query_embeddings=query_embedding,  
        n_results=2  
    )  
    return results['documents'][0]
```

```
# Test function  
print("\nCustom Retrieval Test:\n")  
print(retrieve("Explain machine learning"))
```

Custom Retrieval Test:

```
['Machine Learning is a subset of AI.', 'Supervised learning uses labeled data.']
```

```
queries = [  
    "What is AI?",  
    "Explain deep learning",  
    "What is NLP?",  
    "How does RAG work?"  
]
```

```
for q in queries:  
    print(f"\nQuery: {q}")  
    print("Results:", retrieve(q))
```

Query: What is AI?

Results: ['AI is widely used in healthcare and finance.', 'Artificial Intelligence is the simulation of human intelligence.']

Query: Explain deep learning

Results: ['Deep Learning uses neural networks.', 'Neural networks are inspired by the human brain.']

Query: What is NLP?

Results: ['Natural Language Processing deals with text data.', 'Supervised learning uses labeled data.']

Query: How does RAG work?

Results: ['RAG combines retrieval and generation.', 'Reinforcement learning uses rewards and punishments.']

Start coding or [generate](#) with AI.